



DEALER BEST PRACTICE SERIES

Cost Effective Gasket and Paint
Cleaning Processes

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

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1.0 Introduction

Efficient parts cleaning processes are critical to several facets of CRC operations. Parts cleaning may be considered a non-value adding step, but it represents a substantial portion of labor and expenses in the rebuilding process. The cleaning step is an important segment in the critical path, since the last parts disassembled are the first parts needed in assembly. All other parts must wait in staging until those critical few are cleaned, rebuilt, and cleaned again before the component is ready for assembly.

There are many technologies used in the cleaning process, such as high-pressure cabinet washers, buffing, grinding, blast cleaning, and solvent washing. Ideally, the fastest and least costly process is applied to each part as needed. Automatic cleaning is typically preferred for cost control and consistent quality. Gasket removal and seal surface cleaning is difficult because it often requires extensive manual labor. Paint removal is difficult because:

- Ferrous metals require a caustic solution to remove most paints. This is expensive to buy, and the remaining wastes are expensive to dispose of.
- Non-ferrous metals require very high cost chemicals to strip paint, and these are also very expensive to dispose of in many regions.

This Best Practice will discuss paint and gasket removal without requiring such expensive chemical processes. The processes employed will provide a sufficient level of cleanliness without “over-cleaning.” These new processes will also minimize part damage and processing time/labor.

2.0 Best Practice Description

Parts cleaning is often performed by lesser-skilled employees – those with the least experience in the dealer and/or product. They are typically trained by outgoing personnel from that area, and receive little follow-up unless the rebuild technicians complain about dirty parts. Some rebuild centers will also require technicians to clean parts, with each tech being responsible for his or her own parts. Cleaners typically:

- Grind and scrape all the gaskets off before the cabinet washing. This removal requires time, labor, and effort making the parts more vulnerable to damage and reducing the effectiveness of gaskets and seals. Excess grinding and scraping also creates airborne debris that will spread into the shop and require additional cleanup.
- Grind/scrape all the gaskets off after the cabinet washing, typically long after the parts have cooled and gaskets have hardened, again using lots of time. This excess grinding and scraping creates the same problems mentioned above. In addition, abrasives and debris can contaminate the parts and become incorporated into the rebuilt component, and affect service life.
- Use several methods to remove paint: blast cleaning requires time and facility resources. Grinding/buffing to remove all the paint also requires significant time and labor and spreads airborne debris. Chemical removal consumes caustic, or more expensive chemicals, which add excessive disposal costs to direct expenses. Some operations don’t clean paint beyond the aluminum-safe solution washes. This omission leaves curled paint surfaces on the parts, which:

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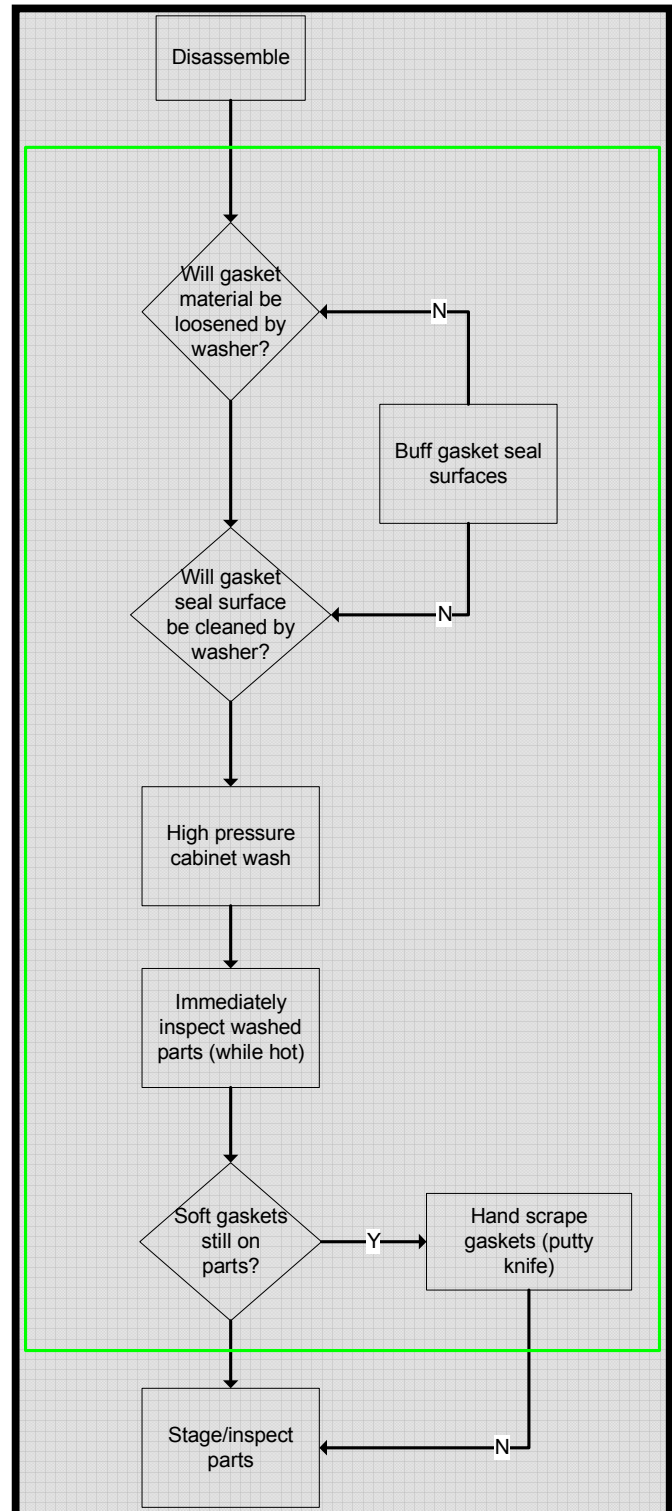
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- Creates an inferior painting surface, resulting in a low-quality product image to the customer.
- Causes paint chips to re-enter the assembly process; contaminating bearing surfaces and blocking lubrication passages. Results can reduce service life or possibly cause early failure.
- Creates a mess in the work bays, with paint chips breaking off the parts and flying around the shop. This adds to shop cleaning efforts and demonstrates an unprofessional image to employees and the customer.

Effective gasket and paint cleaning processes provide balance towards getting parts cleaned to a consistent quality and minimizing the time, labor, and supplies to perform the cleaning. This process follows:

- Gaskets/sealants that soften during the cabinet washer cycle should be quickly scraped off with a putty knife immediately after the wash cycle. The parts should still be hot and the gaskets should be easy to scrape off quickly. Many of the gaskets may have already fallen off the parts during the wash cycle.
- Gaskets that were not softened and subsequently difficult to remove after the wash cycle, should be scraped or ground off before the second wash cycle. This will minimize post-wash grinding and further wash cycles. This is commonly required with gaskets exposed to high heat or those applied with aggressive sealing chemicals. This is common with exhaust gasket and head gasket areas – those exposed to high combustion heat.

Paint & Gasket Removal Process
(Best Practice scope in green box)



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- Once processed by the cabinet washer, painted parts may be:
 - “Buffed” with a powered bench-mounted, or hand-held angle grinder. Although rust/corrosion should be completely removed, only the paint chips need be removed up to the point of where the paint remains adheres to the part. The sharp broken paint edge on the part should only be “feathered” to blend it into the part surface. The blend-line should disappear under a good coat of paint.
 - Blast cleaned with a automatic tumbling cleaner
- Blast clean aggressive corrosion only as needed. This may be performed manually, especially for larger/sensitive parts, or with an automatic tumbling blaster, for other parts. Blast only critical areas and blast to “feather” out paint to minimize time, labor, and supply expense. Always wash parts after blast cleaning due to the amount of abrasive carry-over.

Establish consistent and cost-effective cleaning processes:

- Perform cleaning only with specialized cleaning employees. Do not permit rebuild technician to share in the general cleaning area duties as a rule. Rebuild technicians may only clean in the general cleaning area only under special circumstances and to expedite a repair.
- Establish a training process or course to prepare the cleaning specialist. Include best-in-class visual aids and physical samples. Implement the training consistently with demonstrated qualification standards. Emphasize quality and cost effectiveness. Review performance periodically to assure quality does not drift and process improvements are documented/replicated.
- Establish cleaning equipment maintenance programs for consistent operations. Include schedules, checklists, and reliable supply delivery/inventory systems. Define individual responsibilities for equipment operators and maintainers. Enforce daily/consistently.
- Lead cleaners and disassembly technicians as a team to capitalize on the location, and opportunities to share employees between functional areas. This also allows for better parts reuse and applied failure analysis process control. Parts often must be cleaned for better reuse and AFA decisions, and a team can coordinate these processes with the least effort/ bureaucracy.

3.0 Implementation Steps

1. Document as-is cleaning processes, equipment, and performance.
 - a. Develop practical performance metrics.
 - b. Document subjective observations through definitions and visual aids.
 - c. Account for employee experience and strategy in applying employees to function.
 - d. Include area equipment maintenance programs, roles/responsibilities, and adherence.
2. Define future cleaning strategy based on expected product type, volume, size, etc.
 - a. Plan for typical rebuild volumes as well as expected peak production business cycles.
 - b. Investigate specialized cleaning facilities, equipment, tools and processes.
 - c. Utilize Cat Facility Development and Service Tools Development.
 - d. Visit best-in-class dealers to observe their operations.
3. Apply future strategy to shop layout plans.
 - a. Consider staging location’s effects on product protection and flow.
 - b. Add changes to shop layout drawing.
 - c. Include shop technician teams in the review/application stages.

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- d. Include outside groups providing facility/equipment support functions, as needed.
- e. Use Cat Facility Development and Service Tools Development expertise as required.
4. Develop/document the new cleaning (and disassembly – as needed) strategy/procedure, with new roles and responsibilities.
5. Establish facility/equipment maintenance strategies as needed.
6. Establish adjusted repair/rebuild time requirement targets as needed.
7. Establish adjusted cleaning (and disassembly) group leadership as needed.
8. Present strategy to shop employees.
9. Implement the staging processes, layout, and equipment changes into the shop.
 - a. This may occur in phases (as needed).
 - b. Test new processes to establish “best-fit” application for shop.
10. Train employees to use the new procedures. Follow-up/enforce immediately.
11. Review process and support system performance once established.
12. Establish and implement adjustments as needed.

4.0 Benefits

- Increased capacity - These concepts increased cleaning stage throughput for a given amount of labor hours, by specializing and effectively training the employees. Supply control/availability minimizes production delays.
- Reduced cost – Labor hour targets per cleaning segment were reduced as mentioned above. “Over-cleaning” was almost eliminated and consistently controlled with periodic process review. Equipment is better maintained with consistent cleaners and comprehensive maintenance support. Supply control minimizes production interruptions.
- Increased quality – Reduced over-cleaning allowed time for focusing on the tough cleaning areas, and consistent cleaning methods increased the skills applied to those areas.
- Improved image – Effective processes and employees in the cleaning area promote a higher level of professionalism with the employees. A cleaner environment also promotes employee professionalism and this was immediately obvious to all customers visiting the CRC. They turned a typically “dirty corner” into a productive working area.

5.0 Resources Required

- Investment costs vary, depending on cleaning area/system related gaps found and local labor/material costs:
 - Equipment upgrades to provide product quality/consistency.
 - Facility upgrades to provide effective product flow to match new processes.
 - System development to assure facility/equipment reliability.
- Support Equipment – as needed:
 - Equipment organization aids (racks, cabinets).
 - Airborne debris containment (downdraft tables, evacuation systems, etc.).
 - Utility reels for effective facility cleaning efforts.
 - Automatic cleaning equipment (tumbling blasters, cabinet washers, etc.).
- People
 - Establish current cleaning, quality, costs, and environmental concerns with team.
 - Establish process, facility, and equipment improvements through team.
 - Establish team organization/leadership to coordinate/support contingent functions.

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- Training
 - Establish the improvement benefits with the production team.
 - Define and enforce the new process and duties with the production team.
 - Define and enforce new support systems/process/duties with support teams.

6.0 Supporting Attachments / References

None.

7.0 Related Best Practices

0107-4.5-1060	-CRC Parts Buffer Enhancement
0207-4.5-1063	-CRC Material Transport Strategy
0107-4.4-1061	-CRC Parts Blasting Enhancement
0207-4.4-1066	-CRC Proper Lighting Provides Effective Working Environment

8.0 Acknowledgements

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